

From square root of -1 to square root of multivector (MV) in Clifford Algebras $n \leq 3$

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2 Square root of $Cl_{3,0}$ multivector

3 Examples and implementation

4 Conclusions

Problem formulation

For **ANY** given multivector $B = b_0 + \mathbf{b} + \mathcal{B} + b_{123}I$ of real CA of $n = 3$, where b_0 , $\mathbf{b}$, $\mathcal{B}$, $b_{123}I$ denotes the scalar, the vector, the bivector and the pseudoscalar, respectively, find **ALL** multivectors

$$\begin{aligned} A &= a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 + a_{12}\mathbf{e}_{12} + a_{23}\mathbf{e}_{23} + a_{13}\mathbf{e}_{13} + a_{123}I \\ &\equiv a_0 + \mathbf{a} + \mathcal{A} + a_{123}I, \end{aligned}$$

where a_i , a_{ij} and a_{123} are the real coefficients and where $\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3$ and $\mathcal{A} = a_{12}\mathbf{e}_{12} + a_{23}\mathbf{e}_{23} + a_{13}\mathbf{e}_{13}$ is, respectively, the vector and the bivector, which satisfy

$$B = AA \equiv A^2 = b_0 + \mathbf{b} + \mathcal{B} + b_{123}I. \quad \text{Root equation}$$

The MV **A** is called a square root of **B**.

Summary of results

Strict algorithm has been worked out

which finds ALL square roots in radicals. In particular we state that for $n \leq 3$ the given MV may have

- **no roots** (with some exceptions),
- **isolated roots** (up to 16 roots for $Cl_{2,1}$),
- **continuum of roots** (except for $n = 1$)

Files

We demonstrate the implementation of the algorithm in *Mathematica* package, which may be download under licence GPL-3.0 from

<https://github.com/ArturasAcus/GeometricAlgebra>

Methods used

Application of 8-periodicity table

Investigations of conditions for existence of square root of -1 in the Clifford-Fourier (CF) transforms and CA based wavelet theory

- $p + q \leq 4$ E. Hitzer, R. Abłamowicz, "Geometric roots of -1 in Clifford algebras $Cl_{p,q}$ with $p + q \leq 4$ ", *Adv. Appl. Clifford Algebras*, **21**(1):121–144, 2011.
- $p + q = 5; 7$; $p - q = 3(\text{mod}4)$. E. Hitzer, J. Helmstetter, R. Abłamowicz, "Square roots of -1 in real Clifford algebras", in E. Hitzer, S. J. Sangwine (Eds.), *Quaternion and Clifford-Fourier Transforms and Wavelets*, pp. 123–153, Springer, Basel, 2012. Here the
- Wensheng Cao, Quadratic equation in split quaternions Clifford algebra, arXiv:1905.08153v3 [math.RA] 11 May 2020

Uptill now, all methods to calculate the square root algebraically were based either on matrix isomorphism (8-periodicity Bott table) or started from the root equation.

Methods used

The method used in the present report

In this report the algebraic and numerical analysis based on root equation $A^2 = B$ was applied. The preliminary results are presented in

- A. Dargys, A. Acus, "Square root of a multivector in 3D Clifford algebras", Nonlinear Analysis: Modelling and Control, Vol. **25**, No. 2, 301320
- A. Acus, A. Dargys, "Square root of a multivector of Clifford algebras in 3D: A game with signs", arXiv:2003.06873

Why square roots of Bott matrices fail?

"...for I cannot believe that anything so ugly as the multiplication of matrices is an essential part of the scheme of nature."

sir Arthur Eddington

Matrix based attempt of square root in $Cl_{3,0}$.

In $Cl_{3,0}$ take the MV $B = \mathbf{e}_1 - 2\mathbf{e}_{23}$. Let us calculate $\sqrt{B} = \sqrt{\mathbf{e}_1 - 2\mathbf{e}_{23}}$, Using isomorphism between $Cl_{3,0}$ and Pauli matrices,

$$\{1, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_{12}, \mathbf{e}_{23}, \mathbf{e}_{13}, \mathbf{e}_{123}\} \cong$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix},$$

one has the matrix representation,

$$\hat{B} = \begin{bmatrix} 0 & 1 - 2i \\ 1 + 2i & 0 \end{bmatrix}$$

Why using matrix representations in GA fails?

Matrix based attempt of square root in $Cl_{3,0}$ (continuation)

To find the square root of matrix $\hat{A}$, Sullivan's formula is used

$$\hat{A} = \sqrt{\hat{B}} = \varepsilon_2 \frac{\hat{B} + \varepsilon_1 \sqrt{\text{Det} \hat{B}} \hat{I}}{\sqrt{\text{Tr} \hat{B} + 2\varepsilon_1 \sqrt{\text{Det} \hat{B}}}},$$

where $\hat{I}$ is the unit matrix and $\varepsilon_i = \pm 1$. For a given $\hat{B}$ and $\varepsilon_1 = \varepsilon_1 = 1$ one can calculate the first root the square which after transformation back to multivector is

$$A = \sqrt{1 + i/2} + \frac{1}{\sqrt{4 + 2i}} \mathbf{e}_1 + \frac{2}{\sqrt{4 + 2i}} \mathbf{e}_{23}.$$

The square of A is $A^2 = \frac{1}{5}(2 + 4i) + \mathbf{e}_1 - 2\mathbf{e}_{23} + \frac{1}{5}(4 - 2i)\mathbf{e}_{123}$. It is not equal to initial MV: $A^2 \neq B = \mathbf{e}_1 - 2\mathbf{e}_{23}$. Why? Correct solution

Why using matrix representations in GA fails?

Matrix based attempt of square root in $Cl_{3,0}$ (continuation)

Possible explanation can be found P. Lounesto book **Clifford Algebras and Spinors**, *Cambridge University Press, Cambridge, 2001*. In p. 54 he explicitly states:

“The essential difference between the Clifford algebra Cl_3 and its matrix image $Mat(2, C)$ is that in the Clifford algebra Cl_3 we will, in its definition, **distinguish a particular subspace, the vector space R^3** in which the square of a vector equals its length squared, that is, $r^2 = |r|^2$. No such distinguished subspace has been singled out in the definition of the matrix algebra $Mat(2, C)$. Instead, we have chosen the traceless Hermitian matrices to represent R^3 , and thereby added extra structure to $Mat(2, C)$.”

Exercise: define blade and versor in terms of matrices.

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The algebraic method

Use Groebner basis and variable replacements to solve components.

Algebraic formulation

Rewrite the MV A as a sum of duals and then equate the coefficients at different basis elements of root equation $A^2 = B$

$$A = s + \mathbf{v} + (S + \mathbf{V})I, \quad s, S \text{ are scalars and } \mathbf{v}, \mathbf{V} \text{ are vectors}$$

$$B = b_0 + b_1\mathbf{e}_1 + b_2\mathbf{e}_2 + b_3\mathbf{e}_3 + b_{12}\mathbf{e}_{12} + b_{13}\mathbf{e}_{13} + b_{23}\mathbf{e}_{23} + b_{123}\mathbf{e}_{123}$$

s, S are scalars. $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + v_3\mathbf{e}_3$ and $\mathbf{V} = V_1\mathbf{e}_1 + V_2\mathbf{e}_2 + V_3\mathbf{e}_3$.

$$b_0 = s^2 - S^2 + \mathbf{v}^2 - \mathbf{V}^2, \quad b_{123} = 2(sS + \mathbf{v} \cdot \mathbf{V}), \quad (1)$$

$$b_1 = 2(sv_1 - SV_1), \quad b_{23} = 2(sV_1 + Sv_1), \quad (2)$$

$$b_2 = 2(sv_2 - SV_2), \quad b_{13} = 2(sV_2 + Sv_2), \quad (3)$$

$$b_3 = 2(sv_3 - SV_3), \quad b_{12} = 2(sV_3 + Sv_3). \quad (4)$$

The algebraic method

Step 1: Solve for $\mathbf{v}$ and $\mathbf{V}$

Note that Eqs (2)–(4) are linear with respect to coefficients (v_1, v_2, v_3) and (V_1, V_2, V_3) . Therefore, the coefficients are single-valued:

$$v_1 = \frac{b_1 s + b_{23} S}{2(s^2 + S^2)}, \quad v_2 = \frac{b_2 s - b_{13} S}{2(s^2 + S^2)}, \quad v_3 = \frac{b_3 s + b_{12} S}{2(s^2 + S^2)}, \quad (5)$$

$$V_1 = \frac{b_{23} s - b_1 S}{2(s^2 + S^2)}, \quad V_2 = -\frac{b_{13} s + b_2 S}{2(s^2 + S^2)}, \quad V_3 = \frac{b_{12} s - b_3 S}{2(s^2 + S^2)}. \quad (6)$$

Substituting v_i and V_i into original dual system (1) we have two coupled algebraic equations of order 4 for two unknowns s and S

$$\begin{aligned} 4(b_0 - s^2 + S^2)(s^2 + S^2)^2 = &+ (b_1 s + b_{23} S)^2 + (b_2 s - b_{13} S)^2 + (b_3 s + b_{12} S)^2 \\ &- (b_{23} s - b_1 S)^2 - (b_{13} s + b_2 S)^2 - (b_{12} s - b_3 S)^2, \\ 2(b_{123} - 2sS)(s^2 + S^2)^2 = &+ (b_1 s + b_{23} S)(b_{23} s - b_1 S) - (b_2 s - b_{13} S) \\ &\times (b_{13} s + b_2 S) + (b_3 s + b_{12} S)(b_{12} s - b_3 S). \end{aligned} \quad (7)$$

The algebraic method

Step 2: variable substitution

The coupled system of equations can be solved using substitution

$$s S = t, \quad \frac{1}{2}(-s^2 + S^2) = T, \quad (8)$$

which reduces the system (7) to

$$\begin{aligned} (b_0 + 4T)(4t - b_{123}) - b_I/2 &= 0, \\ b_S - (b_0 - b_{123} + 4T + 4t)(b_0 + b_{123} + 4T - 4t) &= 0, \end{aligned} \quad (9)$$

with a coordinate-free abbreviations b_S and b_I ,

$$\begin{aligned} b_S &= \langle \widehat{B\bar{B}} \rangle_0 = b_0^2 - b_1^2 - b_2^2 - b_3^2 + b_{12}^2 + b_{13}^2 + b_{23}^2 - b_{123}^2, \\ b_I &= \langle \widehat{B\bar{B}I} \rangle_0 = 2b_3b_{12} - 2b_2b_{13} + 2b_1b_{23} - 2b_0b_{123}. \end{aligned} \quad (10)$$

The algebraic method

Step 3: the answer

$$s_{1,2} = \pm \sqrt{-T + \sqrt{T^2 + t^2}}, \quad S_{1,2} = \pm \frac{t}{\sqrt{-T + \sqrt{T^2 + t^2}}}, \quad \text{where}$$

$$\begin{cases} t_{1,2} = \frac{1}{4} \left(b_{123} \pm \frac{1}{\sqrt{2}} \sqrt{-b_S + \sqrt{D}} \right), & T_{1,2} = \frac{1}{4} \left(\frac{\pm b_I}{\sqrt{2} \sqrt{-b_S + \sqrt{D}}} - b_0 \right), \\ & \text{if } -b_S + \sqrt{D} > 0 \\ t_{1,2} = \frac{1}{4} b_{123}, & T_{1,2} = \frac{1}{4} (\pm \sqrt{b_S} - b_0), \quad \text{if } -b_S + \sqrt{D} = 0 \text{ and } b_S > 0 \end{cases}$$

and

$$b_S = \langle \widehat{B\bar{B}} \rangle_0 = \begin{cases} b_0^2 - b_1^2 - b_2^2 - b_3^2 + b_{12}^2 + b_{13}^2 + b_{23}^2 - b_{123}^2, & \text{for } Cl_{3,0} \\ b_0^2 - b_1^2 + b_2^2 + b_3^2 - b_{12}^2 - b_{13}^2 + b_{23}^2 - b_{123}^2, & \text{for } Cl_{1,2} \end{cases}$$

$$b_I = \langle \widehat{B\bar{B}I} \rangle_0 = 2b_3b_{12} - 2b_2b_{13} + 2b_1b_{23} - 2b_0b_{123},$$

$$D = b_S^2 + b_I^2, \quad D \text{ denotes the determinant of } B.$$

The algebraic method

The case $s^2 + S^2 \neq 0$

Only in this case the coefficients of $\mathbf{v}$ and $\mathbf{V}$ are computed as in (5) and (6) since equations contain division by $(s^2 + S^2)$.

The case $s^2 + S^2 = 0$

A solution exists only when $b_1 = b_2 = b_3 = b_{12} = b_{13} = b_{23} = 0$. The six coefficients of vectors $\mathbf{v}$ and $\mathbf{V}$ then are related by two equations

$$b_0 = \mathbf{v}^2 - \mathbf{V}^2, \quad b_{123} = 2(\mathbf{v} \cdot \mathbf{V}),$$

therefore we are left with **four** free parameters, provided that the above system allows real solution for some real values of free parameters. The equations yields clear geometric interpretation!

The algebraic method

The case $s^2 + S^2 = 0$

When solved for v_1 and V_1 the solution is

$$v_1 = \mp \frac{c_1}{\sqrt{2}}, \quad V_1 = \pm \frac{1}{c_1} \frac{-b_{123} + 2(v_2 V_2 + v_3 V_3)}{\sqrt{2}}, \quad \text{with}$$

$$c_1 = \left(\pm \sqrt{(b_0 - v_2^2 - v_3^2 + V_2^2 + V_3^2)^2 + (b_{123} - 2(v_2 V_2 + v_3 V_3))^2} + (b_0 - v_2^2 - v_3^2 + V_2^2 + V_3^2) \right)^{\frac{1}{2}}$$

where vector components v_2, v_3 and V_2, V_3 are arbitrary and may be treated as free parameters.

Algorithm and results

The algorithm for $Cl_{3,0}$

Input: $B = b_0 + b_1\mathbf{e}_1 + b_2\mathbf{e}_2 + b_3\mathbf{e}_3 + b_{12}\mathbf{e}_{12} + b_{13}\mathbf{e}_{13} + b_{23}\mathbf{e}_{23} + b_{123}\mathbf{e}_{123}$

Output: $A = s + \mathbf{v} + (S + \mathbf{V})I$ with the property $A^2 = B$ or $\emptyset$

Initialization: $b_S = b_0^2 - b_1^2 - b_2^2 - b_3^2 + b_{12}^2 + b_{13}^2 + b_{23}^2 - b_{123}^2$;

$b_I = 2b_3b_{12} - 2b_2b_{13} + 2b_1b_{23} - 2b_0b_{123}$; $D = b_S^2 + b_I^2$;

/ compute all (t_i, T_i) pairs* **/*

if $-b_S + \sqrt{D} > 0$ **then**

$t_{1,2} = \frac{1}{4} \left(b_{123} \pm \frac{1}{\sqrt{2}} \sqrt{-b_S + \sqrt{D}} \right)$, $T_{1,2} = \frac{1}{4} \left(\frac{\pm b_I}{\sqrt{2} \sqrt{-b_S + \sqrt{D}}} - b_0 \right)$;

else if $-b_S + \sqrt{D} = 0$ && $b_S > 0$ **then**

$t_{1,2} = \frac{1}{4} b_{123}$, $T_{1,2} = \frac{1}{4} (\pm \sqrt{b_S} - b_0)$; */* degenerate case */*

else

return $\emptyset$; */* never happens */*

end

Algorithm and results

The algorithm for $Cl_{3,0}$ (continuation I)

```

/* for each  $(t_i, T_i)$  pair (where  $i = 1, 2 = (+, -)$ ) find corresponding  $(s_i, S_i)$ 
   pair */
foreach  $(t_i, T_i)$  do
     $s_i = \pm \sqrt{-T_i + \sqrt{T_i^2 + t_i^2}}, \quad S_i = \pm \frac{t_i}{\sqrt{-T_i + \sqrt{T_i^2 + t_i^2}}},$ 
end
/* index  $i = 1, 2 = (+, -)$  match two values */

```

Algorithm and results

The algorithm for $Cl_{3,0}$ (continuation II)

```

/* for each  $(s_i, S_i)$  pair compute corresponding  $v_{i_k}, V_{i_k}$  */
foreach  $(s_i, S_i)$  do
  if  $s^2 + S^2 \neq 0$  then
     $v_{i_1} = \frac{b_1 s + b_{23} S}{2(s^2 + S^2)}, v_{i_2} = \frac{b_2 s - b_{13} S}{2(s^2 + S^2)}, v_{i_3} = \frac{b_3 s + b_{12} S}{2(s^2 + S^2)}, V_{i_1} = \frac{b_{23} s - b_1 S}{2(s^2 + S^2)}, V_{i_2} = -\frac{b_{13} s + b_2 S}{2(s^2 + S^2)}, V_{i_3} = \frac{b_{12} s - b_3 S}{2(s^2 + S^2)}.$ 
    return  $A \leftarrow (s_i, S_i, v_{i_k}, V_{i_k});$  /* isolated root */
  else
    if  $b_1 = b_2 = b_3 = b_{12} = b_{13} = b_{23} = 0;$  /*  $s^2 + S^2 = 0$  case */
    then
      solve any two of  $v_{i_k}, V_{i_j}$  from  $\begin{cases} b_0 = v^2 - v^2, \\ b_{123} = 2(v \cdot v) \end{cases}$ 
      return  $A \leftarrow (s_i, S_i, v_{i_k}, V_{i_j});$  /* continuum root */
    else
      return  $A \leftarrow \emptyset;$  /* no root */
    end
  end
end

```

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Examples

The square root of $B = \mathbf{e}_1 - 2\mathbf{e}_{23}$ of $Cl_{3,0}$

The only nonzero coefficients are $b_1 = 1$ and $b_{23} = -2$. We have $b_I = -4$, $b_S = 3$ and $D = 25$. The values for t and T are $t_{1,2} = (\frac{1}{4}, -\frac{1}{4})$ and $T_{1,2} = (-\frac{1}{2}, \frac{1}{2})$. Finally the real solutions for s and S are

$$(s_{1,2} = \mp \frac{1}{2}c_1, S_{1,2} = \pm \frac{1}{2}c_2) \quad \text{and} \quad (s_{3,4} = \pm \frac{1}{2}c_2, S_{3,4} = \pm \frac{1}{2}c_1),$$

where $c_1 = \sqrt{-2 + \sqrt{5}}$ and $c_2 = \sqrt{2 + \sqrt{5}}$. Since $s \neq S$ we have the following four sets of non-zero coefficients:

$$\begin{aligned} & (s_1 = -\frac{1}{2}c_1, S_1 = \frac{1}{2}c_2, v_{11} = -\frac{1}{2}c_2, V_{11} = -\frac{1}{2}c_1), \\ & (s_2 = \frac{1}{2}c_1, S_2 = -\frac{1}{2}c_2, v_{21} = \frac{1}{2}c_2, V_{21} = \frac{1}{2}c_1), \\ & (s_3 = \frac{1}{2}c_2, S_3 = \frac{1}{2}c_1, v_{31} = \frac{1}{2}c_2, V_{31} = -\frac{1}{2}c_1), \\ & (s_4 = -\frac{1}{2}c_2, S_4 = -\frac{1}{2}c_1, v_{41} = -\frac{1}{2}c_1, V_{41} = \frac{1}{2}c_2). \end{aligned}$$

The remaining coefficients are equal to zero, $v_{i2} = v_{i3} = V_{i2} = V_{i3} = 0$.

Examples

The square root of $B = \mathbf{e}_1 - 2\mathbf{e}_{23}$ (continuation)

Inserting the coefficients into parametrization we write down four roots,

$$A_{1,2} = \mp \frac{1}{2} c_2 (-2 + \sqrt{5} + \mathbf{e}_1 + (-2 + \sqrt{5})\mathbf{e}_{23} - \mathbf{e}_{123}),$$

$$A_{3,4} = \pm \frac{1}{2} c_1 (2 + \sqrt{5} + \mathbf{e}_1 - (2 + \sqrt{5})\mathbf{e}_{23} + \mathbf{e}_{123}),$$

the squares of which yields $A^2 = \mathbf{e}_1 - 2\mathbf{e}_{23}$. Compare it with the square root of matrix representation (see the slide "Why square roots of Bott matrices fail?").

Examples

The square root of $B = \mathbf{e}_1 + \mathbf{e}_{12}$

This MV represents physically important case of linearly polarized electromagnetic wave in $Cl_{3,0}$. We have $b_1 = 1, b_{12} = 1$. Then ($\mathbf{s}_1 = 0, \mathbf{S}_1 = 0$) and $b_0 = b_{123} = b_I = b_S = 0$. Then after substitution of $\mathbf{s} \rightarrow s_1 = 0$ and $\mathbf{S} \rightarrow S_1 = 0$ into

$b_1 = 2(\mathbf{s}v_1 - \mathbf{S}V_1), \quad b_{12} = 2(\mathbf{s}V_1 + \mathbf{S}v_1)$ we obtain a contradiction $1 = 0$. This means that in this case ($Cl_{3,0}$ algebra) the MV $B = \mathbf{e}_1 + \mathbf{e}_{12}$ has no roots.

Interactive demonstration

Implementation in *Mathematica*

We demonstrate the implementation of the algorithm in *Mathematica* using <https://github.com/ArturasAcus/GeometricAlgebra>.

Interactive notebook Sqrt_ICCA12.nb demonstration for $\sqrt{+1}$, $\sqrt{0}$ and $\sqrt{-1}$ for $Cl_{3,0}$ and for generic MV of $Cl_{2,1}$, including 16 isolated roots.

Any other input for arbitrary MV in $n \leq 3$ algebras is welcome!

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Conclusions

- We provide algebraic formulas (an algorithm) to express the square root of general MV in $n \leq 3$ algebras in radicals.
- MVs can have multiple (up to 16 in case of $Cl_{2,1}$) isolated roots, or conversely there may be no real CA roots at all. The MV in $n > 1$ algebras may also accommodate a number free parameters that bring a continuum of square roots on respective parameter hypersurfaces.
- We implemented the described algorithm in *Mathematica* system and verified it by purely numerical search of square roots using *Mathematica* universal root search algorithm realized in the system function **FindInstance**[].

Thank You

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